

Supply Chain Optimization

& Logistics Analytics

Control Tower Dashboard & Analysis

180,519 Orders | 2015-2018

Agenda

- Business Context & Objectives
- Data Overview & Methodology
- Key Performance Metrics
- Hypothesis Results (H1-H5)
- Inventory & Supplier Analysis
- Predictive ML & Limitations
- Recommendations & Next Steps

Business Context

A global manufacturing company faced rising freight costs, unpredictable lead times, and inventory imbalances. This project analyzes end-to-end supply chain data to identify bottlenecks, score supplier reliability, and optimize logistics operations.

Objectives:

- Calculate OTIF and core logistics KPIs
- Identify delay-prone routes and shipping modes
- Score and rank supplier reliability
- Optimize inventory with ABC/XYZ analysis
- Predict late delivery risk with ML
- Build an interactive Control Tower dashboard

Key Performance Metrics

Total Orders	180,519
OTD Rate	42.7%
Late Delivery Rate	57.3%
Avg Lead Time	3.5 days
Avg Shipping Delay	0.57 days
Total Revenue	\$36.8M
Total Profit	\$5.9M
Lost Revenue	\$1.57M
Est. Inventory Turnover	12.2x/year

Hypothesis Results

H1: Expedited > Standard shipping

Result: Rejected. Standard Class 60.2% on-time vs First Class 0.0% (data anomaly)

H2: Ocean routes highest variance

Result: Partially confirmed. Central Africa worst (60.7% late)

H3: 20% suppliers cause 80% delays

Result: Confirmed. Top 20% of origins = 71% of late orders

H4: Late delivery = lower margins

Result: Not confirmed. \$33.08 on-time vs \$32.67 late (negligible)

H5: Q4 seasonal peak

Result: Not confirmed. Q1 highest (\$9.47M), Q4 lowest (\$8.47M)

Inventory & Supplier Insights

ABC Classification:

6 products drive 70% of revenue (A-items). 108 products are C-items (<10%).

XYZ Demand Volatility:

115 products have stable demand (X). 3 have moderate variance (Y). 0 volatile (Z).

Supplier Performance:

3,665 origin locations scored on composite reliability. Top origins have on-time rates >90%. Worst origins have late rates >90%.Best lead-time consistency: Villeurbanne (std 0.67 days). Worst: Makati (std 1.54).

Department Profitability:

Fan Shop (\$2.9M profit, 66K orders) is the most profitable department.

Predictive ML: Late Delivery Risk

Model: XGBoost Classifier with hyperparameter tuning.

Features: 32 static order features (no leaky post-hoc variables).

Split: Time-based (80% train before Apr 2017, 20% test after).

Results:

- Test ROC-AUC: 0.525 (near random)
- Best CV ROC-AUC: 0.589 (tuning did not generalize)
- Precision: 56.4% | Recall: 55.6%
- No dominant feature -- all importances < 0.06
- Geography > shipping mode in impact (0.094 vs 0.0)

Key Insight:

Static order features alone cannot predict future delivery delays. Non-stationarity in logistics data requires real-time feeds (weather, port congestion, carrier status) for production-grade prediction.

Recommendations

- Prioritize Standard Class shipping (60.2% on-time)
- Establish regional distribution hubs for Central Africa & South Asia
- Audit and improve top 20% origins causing 71% of late orders
- Integrate real-time logistics data for ML production deployment
- Tighten inventory control on 6 A-items (70% of revenue)
- Investigate \$1.57M in lost revenue from canceled/fraud orders
- Implement supplier scoring dashboard for data-driven decisions